

Week 1 - Day 2b Supplementary: Lists, While Loops and Dictionaries

This section contains extra material related to lists, while loops and dictionaries.

Exercise 1: Element Swapping []

Write code that swaps the first and third element of a list. For example, for the list `lst = [3, 4, 5, 6, 7]`, the output should be `[5, 4, 3, 6, 7]`.

```
# TODO: Your code here
```

Exercise 2: World Cup []

The World cup happens every fourth year. For example, the World cup took place in 2010, 2014 and 2018. Write a program that finds when the next World cup will be, given the current year.

Example: if the current year is 2025, then the answer is 2026. If the current year is 2022, then the answer is 2022.

```
# TODO: Your code here
```

Exercise 3: Country Abbreviation []

You have created a new country. Now you must decide what the two letter abbreviation of the country should be. These two letters must occur next to each other in the name of your country. Write a program that finds all the different two letter combinations, given the name of your country.

Example: if `country_name` is `addisababa`, then the answer is `["ad", "dd", "di", "is", "sa", "ab", "ba"]`. Note that `ab` and `ba` occur multiple times in the name, but should only occur once in the answer.

```
# TODO: Your code here
```

Exercise 4: Matrices []

Remember that you can put lists inside of lists to create a matrix.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

4.1 Define a list of lists `M` representing the matrix using nested lists

```
# TODO: Your code here
```

4.2 Now, write a program that changes all zeros to ones and all ones to zeros in a matrix.

```
# TODO: Your code here
```

Exercise 5: List Equality [] [] []

Define a list `l` and write code that prints `True` if all the elements in the list **are equal** and `False` otherwise.

Hint: You can **multiply** lists with integers and you can **compare** two lists to check if they are equal.

Examples:

- `[1,1,1]` --> `True`
- `[]` --> `True`
- `[1,2]` --> `False`
- `["salaam", "salaam"]` --> `True`

```
# TODO: Your code here
```

Exercise 6: Taxi Fare [] [] []

A group of friends will take taxis to a restaurant. There are $n+m$ friends in total, n of them have the Feres app and m of them do not. At most 4 people can fit in one taxi. If at least one person in the taxi has the Feres app, then it costs 200 birr, otherwise it costs 300 birr. Your task is to calculate the minimum possible price that the friends have to pay, given the numbers n and m .

Example: if $n=2$ and $m=12$, then the answer is 1000, because the friends can be divided into four taxis where they pay $200+200+300+300$.

```
# TODO: Your code here
```

Exercise 7: Random Names [] [] []

You are given two strings `vowels` and `consonants`. Your task is to generate 10 **random** four letter names, such that the first and third letters are from `consonants`, and the second and third letter are from `vowels`. For example, the program could generate the names `lena`, but not `admas` (because it is too long) or `nils` (because the last letter is not from `vowels`).

`random.randrange(a, b)` is a function to generate a random integer between `a` and `b` (excluding `b` itself, like `range`). To use this function, you first need to import the `random` module by running `import random` in your notebook. Run the code below to generate a random number between 1 and (excluding) 4.

```
import random

vowels = "aeiou"
consonants = "bcdfghjklmnpqrstvwxyz"

# TODO: Your code here
```

Exercise 8: Password Encryption [] []

You want to share the password to your computer with a friend. But first you must encrypt the password, to make sure that nobody else can read it. The encryption you use works as follows:

First take every third letter of your password, starting with the first letter. Next, take every third letter starting with the second letter, and finally every third letter starting with the third letter. In other words, if the password is represented by the string $s_0 s_1 s_2 s_3 \dots s_{n-1}$, then the encrypted string will be $s_0 s_3 s_6 \dots s_1 s_4 s_7 \dots s_2 s_5 s_8 \dots$.

Your friend must now decrypt the string that you provided. Write a program that finds the original string, given the encrypted string.

Example: if `encrypted_string` is `psrawdso`, then the answer is `password`.

```
encrypted_string = "cgtaoorutnnalis"

# TODO: Your code here
```